

The value-growth factor picks up the differences in returns between the value stock and the growth stock and is a long-short portfolio (in jargon it is *a zero-cost mimicking portfolio*). The weight of exposure to value-growth, which is 0.05, is chosen by the investor.

The desired portfolio is a combination of the market weights now adjusted for the value-growth exposure. Note that the optimal portfolio is not a market-weighted portfolio; it overweights value and is tilted away from growth to achieve the desired exposure. In practice, the desired portfolio may have negative weights if the loading on value-growth is high enough, which would require dynamic leverage of the overall portfolio.

Now suppose that the value stock has done well and its new, higher price makes it Neutral. “New Neutral” (formerly Value) moves from a market capitalization weight of 0.20 to 0.40. The old Neutral stock has declined in price from a 50% weight to 30%, making it “New Value.” The Growth stock retains its original characterization—it’s still a growth stock and therefore keeps its original weighting of 30%. The value-growth factor changes: it drops the old value stock (which is now neutral) and brings on board the new value stock (which was previously neutral). Now we have:

	Market Portfolio	Value-Growth Factor	Benchmark Portfolio
New Neutral (Formerly Value)	0.40		0.40
New Value (Formerly Neutral)	0.30	0.05	0.35
Growth	0.30	−0.05	0.25
Sum Weights	1.00	0.00	1.00

The optimal portfolio reflects the change. In a pure market portfolio benchmark, there is no rebalancing to take account of the dynamic value-growth effect. The factor benchmark does this.

Although highly stylized, our simple example illustrates three key points about factor benchmarks:

1. The desired exposure to the value-growth factor is set by the investor (here, a weighting of 5%).
2. The benchmark portfolio is no longer market-weighted and instead reflects the value-growth exposure desired by the investor.
3. The benchmark portfolio is “passive” in the sense that it is based on a set of systematic rules but is “dynamic” in the sense that its composition changes over time.

All the major index companies put out nonmarket weighted indexes, as do some asset management companies. Arnott, Hsu, and Moore (2005), for example,